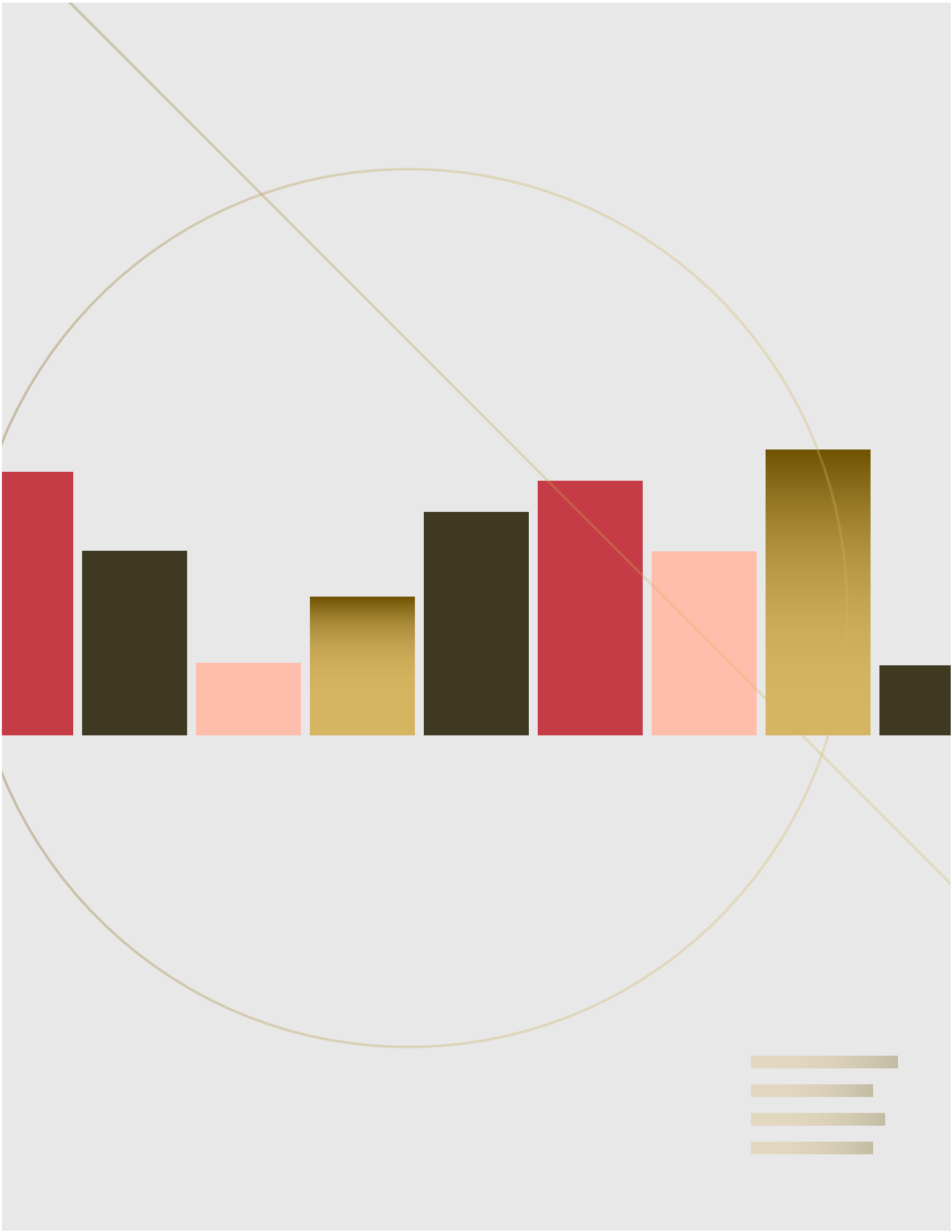




FULL STATISTICS COURSE:

$$z(0.975) = 1.96$$

$$z(0.95) = 1.644854$$

$$z(0.9) = 1.28$$

$$z(0.99) = 2.32$$

Lecture 2:

Gaussian:

$$x \sim \mathcal{N}(\mu, \sigma^2) -$$

$$E[x] = \mu.$$

$$\text{Var}(x) = \sigma^2 > 0.$$

Two Imp Probability tools:

All of probability is relevant to statistics & machine learning.

We replace Expectations by averages!
Why?

Law of Large Numbers:

$$\text{LLN: } \bar{x}_n = \frac{\sum x_i}{n}$$

by a.s.
 $n \rightarrow \infty$

If this happens
all happen.
How close?

$$\mu[E[x_i]]$$

CLT:

$$\frac{\sqrt{n}(\bar{x}_n - \mu)}{\sigma} \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, 1).$$

$$\Rightarrow \sqrt{n}(\bar{x}_n - \mu) \xrightarrow[n \rightarrow \infty]{(d)} \mathcal{N}(0, \sigma^2)$$

$$|\bar{X}_n - \mu| = \frac{1}{\log(\log(\log(n)))}$$

Not useful =

A standard Gaussian is a no. b/w -3 and 3 with overwhelming probability.

Rule of thumb:
 $n \geq 30$

$$\therefore \sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \sim N(0, 1)$$

$(-3, 3)$

Stochastically bounded

I don't want
3, I want 95%.

We want to see how to use it =

$$\therefore |\bar{X}_n - \mu| \leq \frac{3\sigma}{\sqrt{n}} \text{ with}$$

overwhelming probability.

1.2.5

Properties of Gaussian:

Modes of Convergence:

① LLN

② CLT

③

Three types of Convergence:

$T_n \rightarrow$ A sequence of r.v.s.

$T \rightarrow$ A r.v. or deterministic.

① Almost surely: (a.s.)

$$T_n \xrightarrow[n \rightarrow \infty]{\text{a.s.}} T \quad \text{iff} \quad \mathbb{P} \left[\left\{ \omega : \begin{matrix} T_n(\omega) \xrightarrow[n \rightarrow \infty]{} \\ T(\omega) \end{matrix} \right\} \right] = 1.$$

In probability:

$$T_n \xrightarrow[n \rightarrow \infty]{p} T \quad \text{iff} \quad \mathbb{P} \left[|T_n - T| \geq \epsilon \right] \xrightarrow[n \rightarrow \infty]{} 0$$

$\forall \epsilon > 0$

Meaning of almost surely:

if I flip 2 coins,

→ If I get 2 times the same result
my random variable $X = 3$.

otherwise 0 & $X \neq 3$.

Almost surely means:

$$T(\omega: HH, HT) \rightarrow \left[\left\{ T_n(\omega) \xrightarrow{n \rightarrow \infty} T(\omega) \right\} \right] = 1.$$

If I have twice the same
result,
my random variable is $\frac{1}{n}$

other than

↑ 0 Probability $X_n = 1 - \frac{1}{n}$.

↑ An aspect
is
converging.

I want my Random Variables to
converge no matter what the outcome really is.

$T_n \rightarrow T$, in a.s., and probability
we can discuss.

CONVERGENCE IN DISTRIBUTION:
(whole other animal).

$$T_n \xrightarrow[n \rightarrow \infty]{(d)} T \quad \text{iff} \quad E[f(T_n)] \xrightarrow[n \rightarrow \infty]{} E[f(T)],$$

for all continuous and
bounded f .

This simply means
if $f(T_n)$ is b/w a and b -
 $f(T)$ is b/w a and b

CONVERGENCE IN DISTRIBUTION
IMPLIES convergence of probability if
the limit has a density, (eg Gaussian)

$$T_n \xrightarrow{(d)} T \quad \Rightarrow \quad P(a \leq T_n \leq b) \xrightarrow[n \rightarrow \infty]{} P(a \leq T \leq b).$$

Lecture 4.

4.2) Statistics, Estimators, Consistency and Normality.

Statistic:

Any measurable fn. of the sample.

Depend only on your data

fn. of data

Estimator of θ :

Statistic whose exp does not depend on θ .

$(\bar{X}_n, \max_{1 \leq i \leq n} X_i, \log(1 + X_n))$.
sample variance

(which makes it a random variable)

$\hat{\theta}_n \xrightarrow[n \rightarrow \infty]{\text{p.a.s.}} \theta$ w.r.t. P_θ .
weakly Strongly

Asymptotically Normal if:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow[n \rightarrow \infty]{(d)} N(0, \sigma^2)$$

$\sigma^2 \rightarrow$ Asymptotic variance

inf

$$f \in \mathcal{H} \quad \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2$$

Anything inf or sup other than $\rightarrow$ Non-parametric statistics

As i collect more and more data, I should have a better idea, what distribution generated that data.

If our model is both well specified and identifiable.

→ Estimator is any statistic that does not depend on data.

4.3. Bias of an Estimator

$\hat{\theta}_n$ is a random variable
and it has to be centered about what?

↳ Expectation is the centering parameter.

$$\text{bias}(\hat{\theta}_n) = E[\hat{\theta}_n] - \theta.$$

This tells us
how far my random
variable is from the
parameter I am
trying to estimate.

Estimator
Also a Random
Variable
(unknown
parameter).

has all properties of a random
variable like mean and variance.

If $\text{bias}(\hat{\theta}_n) = 0 \Rightarrow$ Unbiased.

Imp: Even before consistency,
we need to check for bias.

But, trying to enforce unbiasedness at
all costs, may lead to bad estimator.

There are estimators that are unbiased
but have a lot of variability.

STATISTIC:

Examples: $\bar{X}_n$, $\max_i X_i$, $X_1 + \log(1 + (X_n)!) .$

One more $\Rightarrow$ MLE will also be there being on avg itself.

ESTIMATOR:

2 ways

① Consistent if $\hat{\theta}_n \rightarrow \theta$

① In probability

② Almost surely

- That's actually the dist of my data.

$$\hat{\theta}_n \rightarrow \theta$$

$$P(|\hat{\theta}_n - \theta| > \epsilon) \xrightarrow{n \rightarrow \infty} 0$$

The unknown parameter:

② Asymptotically Normal

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow[n \rightarrow \infty]{(d)} N(0, \sigma^2)$$

This thing satisfies CLT properties.

$\sigma^2 \rightarrow$ Asymptotic variance.

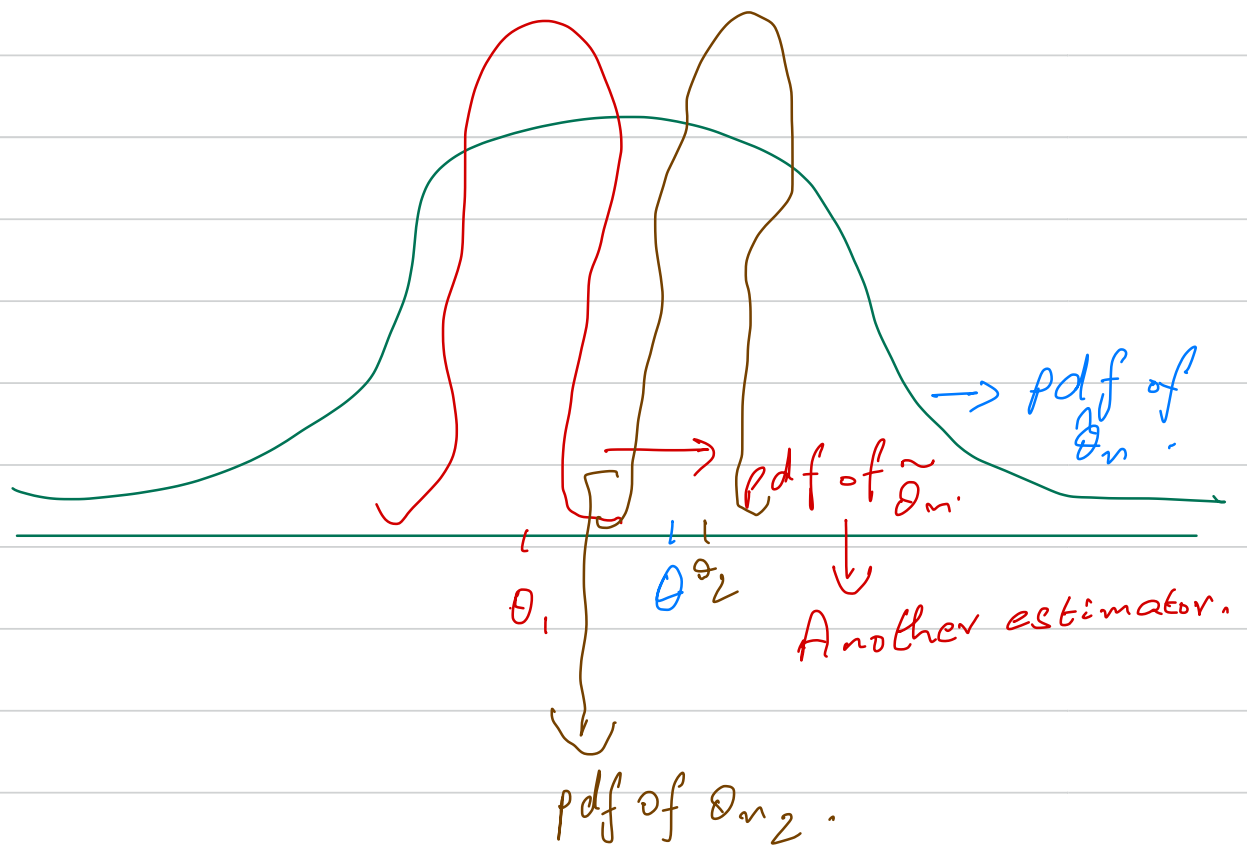
As $n \rightarrow \infty$, I should know what distribution generated that data.

$\Rightarrow$ This will typically be an application of LLN, or CLT, coupled with LLN.

→ Asymptotic variance is not the limit of $\hat{\sigma}_n$.

$$\text{Var}(\hat{\theta}_n) \xrightarrow{n \rightarrow \infty} 0.$$

If my estimator is unbiased, it will look like this:



Out of the three, the best is θ_{n2} because it has low variability and give consistent values close to θ .

Examples: Assume $x_1, \dots, x_n \sim \text{Ber}(p)$

consider the following estimators for p :

$$\Rightarrow \hat{p}_n = \bar{x}_n \quad \text{bias}(\hat{p}_n) = 0, \\ E[\bar{x}_n] = p, \quad \text{var} = \frac{p(1-p)}{n} \rightarrow 0.$$

$$\Rightarrow \hat{p}_n = x_1 \quad \text{bias}(\hat{p}_n) = 0, \\ E[\hat{p}_n] = p \quad \text{var} = p(1-p).$$

$$\Rightarrow \hat{p}_n = \frac{x_1 + x_2}{2} \quad \text{bias}(\hat{p}_n) = 0, \\ E[\hat{p}_n] = p \quad \text{var} = \frac{p(1-p)}{2}$$

$$\Rightarrow \hat{p}_n = \sqrt{I(x_1=1, x_2=1)} \quad \text{bias}[\hat{p}_n] = p^2 - p, \\ E[\hat{p}_n] = p^2 \quad \text{var} =$$

$\Rightarrow$ It is a concave fn.

$\sqrt{I(x_1=1, x_2=1)} \rightarrow$ This is an indicator variable.

Indicator Random Variables $x_1=1, x_2=1$ are iid.

$$E[x_1 x_2] = E[x_1] E[x_2] = p^2. \therefore$$

This is a $\text{Ber}(p^2)$ Random Variable.

$$E[g(x)] = E[x_1 x_2] \therefore \text{this is a Ber R.V.}$$

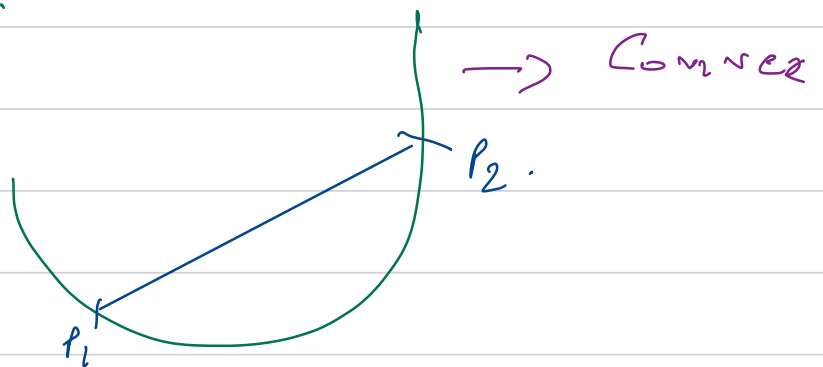
$\Rightarrow E[g(x)] \neq g(E[x])$ in general.

Now, we have something known as

JENSEN'S INEQUALITY:

$\rightarrow$ It says that if we have a fn which is convex, i.e. it looks like this.

$f(\cdot)$.

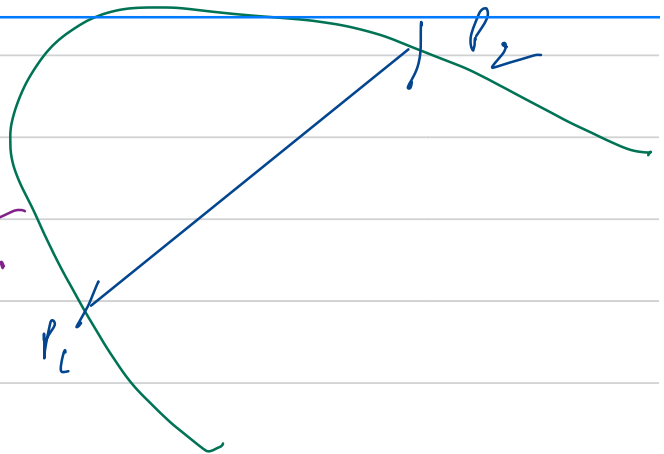


It says that:

$$E[f(x)] \geq f(E[x]).$$

$$E(g(x)) \leq g(E[x]).$$

Concave.



$$E[\sqrt{Z}] < \sqrt{E[Z]} = \rho^2.$$



Strict Inequality for a
concave fn. at

$$\hookrightarrow E[\sqrt{f}] < \sqrt{E[f]} \rightarrow \text{because}$$

at n-point it is flat.

→ If we had something like $|Z|$
things would have been much
harder to compute.

→ The way to remember Jensen, the chord
represents the Expectation, which is always
above the Expectation of the fn. (Genius).

Example:

Quantifying Consistency:

① Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$. Let $\bar{X}_n$ be the estimator given by $\frac{\sum X_i}{n}$.

What is the smallest constant c such that

$$n^c (\bar{X}_n - p) = n^c \left(\frac{\sum X_i}{n} - p \right).$$

Ans. Let $\sigma = \sqrt{p(1-p)}$ denote the common s.d.

of $X_1, \dots, X_n$. By the CLT,

$$\frac{\sqrt{n}}{\sigma} (\bar{X}_n - p) = \frac{\sqrt{n}}{\sigma} \left(\frac{\sum X_i}{n} - p \right) \rightarrow N(0, 1).$$

where the convergence is in distribution. As a result, we see that for $c < 1/2$,

$$n^c (\bar{X}_n - p) = \frac{\sigma}{n^{1/2-c}} \frac{\sqrt{n}}{\sigma} \approx \frac{\sigma}{n^{1/2-c}} N(0, 1) \rightarrow 0$$

almost surely as $n \rightarrow \infty$. Hence, $c = 1/2$ is the smallest value of c such that.

$$n^c (\bar{X}_n - \theta) = n^c \left(\frac{\sum x_i}{n} - \theta \right)$$

does not converge to 0 almost surely as $n \rightarrow \infty$

$\bar{X}_n$ is $\sqrt{n}$ consistent. This means that the estimator $\bar{X}_n$ converges to the true parameter at a relatively fast rate, so this gives us something faster than just consistency.

Q): Imp: Let $x_i \sim U[a, a+1]$, a unknown.

Find an unbiased estimator.

Solⁿ: Let $\bar{X}_n$ denote the estimator.

$$E[\bar{X}_n] = \frac{a+a+1}{2} = a + \frac{1}{2}.$$

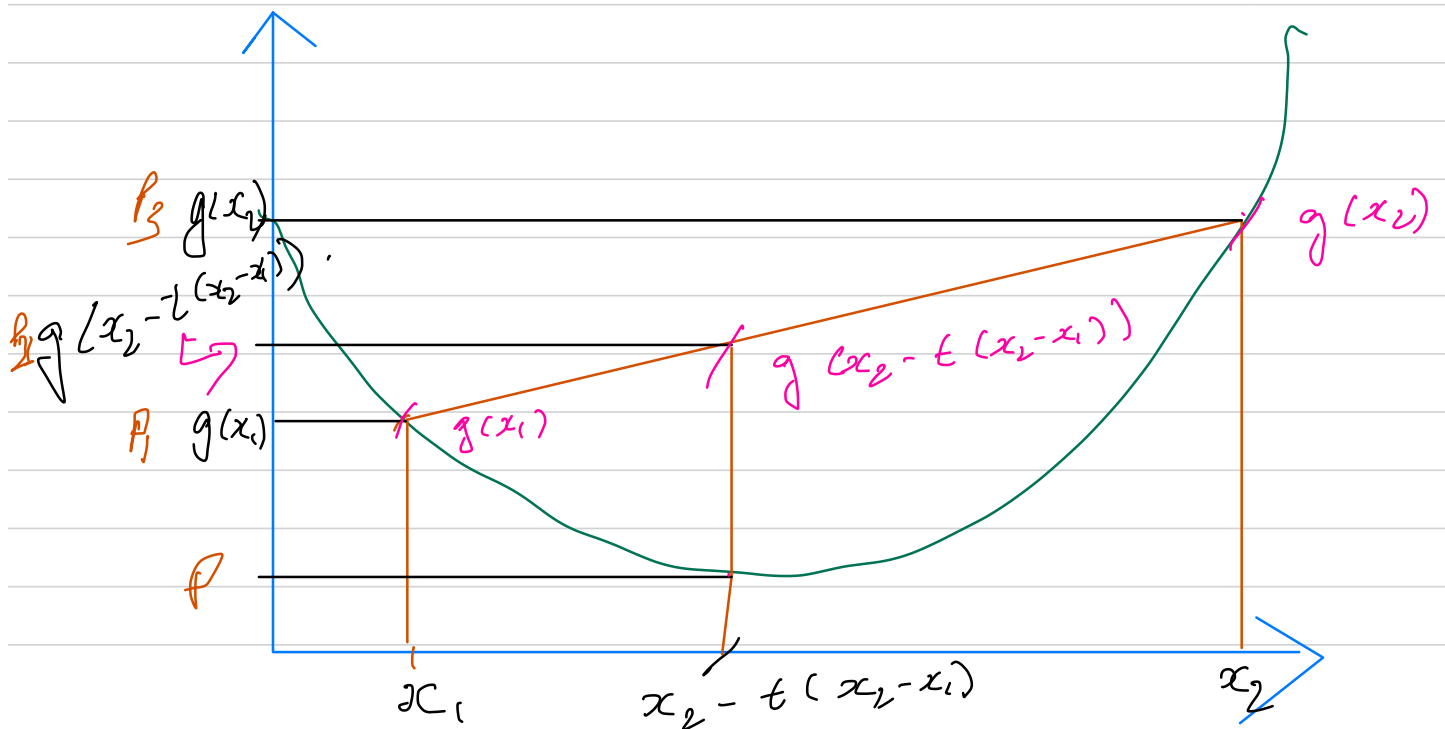
$\therefore \bar{X}_n - 1/2$ is an unbiased estimator w.r.t. a .

JENSEN'S INEQUALITY PROOF:

A fn: $g: \mathbb{R} \rightarrow \mathbb{R}$ is convex if for all pairs $x_1 < x_2$

$$g(tx_1 + (1-t)x_2) \leq tg(x_1) + (1-t)g(x_2).$$

Geometrically, this means for $x_1 \leq x \leq x_2$, the graph of g is below the secant line connecting the points $(x_1, g(x_1))$ and $(x_2, g(x_2))$.



Let's play!

→ It is trying to achieve low bias and low variance at the same time, using

$$\hat{\theta} = E[(\hat{\theta}_n - \theta)^2] = E[(\hat{\theta}_n - E[\theta] + E[\theta] - \theta)^2]$$

$$= E[\hat{\theta}_n^2] + E[\theta^2] - 2 E[\hat{\theta}_n \theta].$$

$$= E[\hat{\theta}_n^2] + E[\theta^2] - 2 E[\hat{\theta}_n] E[\theta].$$

$$= E[\hat{\theta}_n^2] + (E[\theta])^2 - [E[\theta]]^2 + E[\theta^2] - 2 E[\hat{\theta}_n] E[\theta].$$

$$= (E[\hat{\theta}_n - E[\theta]])^2 + \text{var}(\theta).$$

$$\tilde{\theta} = E[(\hat{\theta}_n - \theta)^2] = E[(\hat{\theta}_n - E[\theta])^2] + \text{var}(\theta).$$

$$\tilde{\theta} \neq \Rightarrow \text{var}(\theta) = E[(\hat{\theta}_n - E[\theta])^2] - E[(\hat{\theta}_n - \theta)^2]$$

VARIANCE OF AN ESTIMATOR:

$$\text{Var}(\bar{x}_n) = \text{Var}\left(\frac{\sum x_i}{n}\right) = \frac{1}{n^2} \text{Var}(\sum x_i)$$

$$= \frac{\text{Var}(x_1) + \text{Var}(x_2) + \dots + \text{Var}(x_n)}{n^2}$$

$$= \frac{n}{n^2} \text{Var}(x_1)$$

$$= \frac{1}{n} \text{Var}(x_1).$$

QUADRATIC RISK:

① I have the biases; ② I have the variance

Q) what if they have different bias & variance?

→ I want to minimize this bias-variance trade off!

→ This is done through what's called a quadratic Risk!

INTUITION:

→ First, I want my $\hat{\theta}_n$ close θ .

Let me start with:

$$R(\hat{\theta}_n) = E |\hat{\theta}_n - \theta|$$

I get nothing, let us try squaring it.

$$R(\hat{\theta}_n) = E [|\hat{\theta}_n - \theta|^2] = \text{The rest}$$

↓
One way to make a random variable small is to take its expectation.

→ When I take its square, I am making the random variable $R(\hat{\theta}_n)$'s expectation to be close to zero and variance close to zero.

$$E[(\hat{\theta}_n - \theta)^2] = \text{bias}^2 + \text{variance}.$$

$$\Rightarrow \text{Var}(\theta) = E[(\hat{\theta}_n - \theta)^2] - E[\hat{\theta}_n - E[\theta]]^2.$$

— (1) .

Using Slutsky:

$$E[(\hat{\theta}_n - \hat{\theta}_n)^2] = E[(\hat{\theta}_n - E[\hat{\theta}_n])^2] + \text{Var}(\hat{\theta}_n).$$

$$\Rightarrow \text{Var}(\hat{\theta}_n) = \frac{1}{n} \sigma^2$$

6.8 CONFIDENCE INTERVAL:

→ Just like an estimator depends on my data, my error bars around it will depend on my data.

→ Just like a statistic may depend on θ , my error bars around it may depend on θ .



But it should not.
I want to be able to compute it.

Let $(\bar{X}, (P_\theta)_{\theta \in \Theta})$ be a statistical model on $X_i \sim \text{i.i.d.}$, $\Theta \subseteq \mathbb{R}$. Let $\alpha \in (0, 1)$.

Any Random interval I whose boundaries don't depend on θ
Such

$$P_\theta [I \ni \theta] \geq 1 - \alpha.$$

is called confidence interval $(1 - \alpha)$
for θ .
↑
level.

Meaning of quadratic Risk:

→ The definition of quadratic Risk is the expectation of the square of distance of my Random variable to the point I am trying to estimate.

The way to think about this is: By taking the square root

↑
Qualitative Idea: If I get n in the numerator, as in the example of $U[a, a+1]$, rather than $\frac{1}{12n}$, or something similar

you have made a mistake.

To prove: $E[g(x)] \leq g[E(x)]$.

or equivalently, $g(E[x]) \geq E[g(x)]$

$\Rightarrow g(x_2)$

To prove: $g(x_2 - x_1) \geq g(x_2) - g(x_1)$.

Consider the diagram shown,

Take a point between $x_2 \geq x \geq x_1$.

$$x = x_2 - t(x_2 - x_1) -$$

$$= x_2(1-t) - x_1.$$

$$\Rightarrow g(x) = g(x_2(1-t) - x_1).$$

$\Rightarrow$ For the chord above the graph, let the

Point be called $P \rightarrow$ it's coordinates are

$P_1: (x, y) = (x_2 - t(x_2 - x_1))$ and for the

$P_2: g^{graph}(x, y) = (g(x), g(x_2 - t(x_2 - x_1))).$

$$E[A] E[P_2 - P_1] = x_2 - t(x_2 - x_1) - g(x_1)$$

$$E[B] E[P - P_1] = g(x_2 - t(x_2 - x_1)) - g(x_1).$$

$$\Rightarrow C = A - B = x_2 - t(x_2 - x_1) - g(x_1) - g(x_2 - t(x_2 - x_1)) + g(x_1).$$

$$E[C] = E[P_2 - P].$$

$$E[C] = E[x_2 - t(x_2 - x_1) - g(x_2 - t(x_2 - x_1))]$$

$$= E[x_2 - t(x_2 - x_1)] - E[g(x_2 - t(x_2 - x_1))]$$

$$E[P_3 - P_2] = E[g(x_2)] - E[g(x_2 - t(x_2 - x_1))]$$

$$E[B] - E[C] =$$

$$E[P_2] - E[P] - E[P_3] + E[P] =$$

$$= E[P_2] - E[P_3] \Rightarrow E[C] - E[B] = E[P_3] - E[P_2] \\ = E[g(x_2)] - E[g(x_2 - t(x_2 - x_1))]$$

Now, the inequality can be interpreted as follows:

Let $X \sim \text{Ber}(1-t)$ $\forall$ some $0 \leq t \leq 1$.

$$\Rightarrow g(t(0) + (1-t)(1)) = g(1-t) = g(E[X])$$

$$\text{and } t g(0) + (1-t) g(1) = E[g(X)].$$

$$1 \leq g(1) \text{ for any convex fn. } g(x)$$

$$\therefore \text{For any Bern R.V., } g(E[X]) \leq E[g(X)].$$

JENSEN'S INEQUALITY can be generalised for other R.V.s. For any convex fn. g .

$$g(E[X]) \leq E[g(X)].$$

It is also true for convex fns. and Random Vector $x \sim \mathbb{R}^n$.

Think of α being small, 1%, 2%, 5%, etc.

$$\Rightarrow P_\theta [I \ni \theta] \geq 95\%, 97\%, 99\%.$$

$$\Rightarrow 1 - P_\theta [I \ni \theta] \geq 1 - \alpha.$$

$$\Rightarrow P_\theta [I \ni \theta] \leq \alpha.$$

$\uparrow \rightarrow$ these 2 are equivalent definitions

$$\Rightarrow P_\theta [\theta \in I] \geq 1 - \alpha.$$

$\therefore \theta \in I$ is at least $1 - \alpha$.

According to the defⁿ of α .

Our R.V. should not contain θ .

They way we say the statement is

the probability that our Random Interval contains θ is the same as saying the probability that our Random Interval contains θ , belongs to this C.I.

is at least $1 - \alpha$.

From ②, we have.

$$P[A < \theta < B] \geq 1 - \alpha.$$

$\Rightarrow$ Any random Interval $P(\theta \in (A, B))$
is at least $1 - \alpha$.

$\uparrow$

We can generalise it to asymptotic level $1 - \alpha$.

$$\lim_{n \rightarrow \infty} P_{\theta}[I \ni \theta] \geq 1 - \alpha, \quad \forall \theta \in \underline{\Theta}.$$

$$R_1, \dots, R_n \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(p).$$

$(\mathcal{D}_{0.12}, \text{Ber}(p))_{p \in (0,1)} \rightarrow \text{Statistical model}$

$$\hat{p} = \bar{R}_n$$

Asymptotic Variance

$$\text{CLT: } \sqrt{n} \frac{\bar{R}_n - p}{\sqrt{p(1-p)}} \xrightarrow[n \rightarrow \infty]{\text{d}} N(0, 1).$$

This means:

$$\phi(x) : \text{cdf of } N(0, 1) \quad \phi_n(x) : \text{cdf of } \frac{\sqrt{n}(\bar{R}_n - p)}{\sqrt{p(1-p)}}.$$

CDF

$$\phi_n(x) \approx \phi(x) \rightarrow \text{As } n \text{ becomes large}$$

Defⁿ of Asymptotic Variance:

$$\sqrt{n}(\bar{R}_n - p) \xrightarrow[n \rightarrow \infty]{\text{V. Imp}} N(0, \underline{p(1-p)}).$$

Now,

$$P\left(\frac{\sqrt{n}(\bar{R}_n - p)}{\sqrt{p(1-p)}} \leq x\right) \approx \Phi(x) \quad - (I)$$

We have to calculate $P(|\bar{R}_n - p| \geq x)$.

From (I)

$$P(\bar{R}_n - p \leq x) \approx \Phi\left(0, \frac{p(1-p)}{n}\right).$$

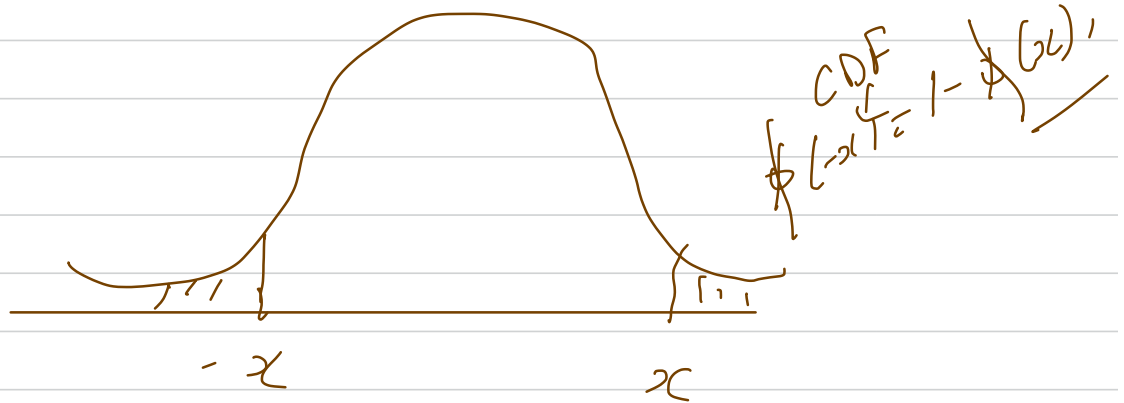
$$\Rightarrow P(-(\bar{R}_n - p) \geq x) \approx \Phi\left(0, \frac{p(1-p)}{n}\right).$$

$$\Rightarrow P(-(\bar{R}_n - p) \geq x) \approx 1 - \Phi\left(0, \frac{p(1-p)}{n}\right).$$

$$\Rightarrow P(|\bar{R}_n - p| \geq x) \approx 1 - 2 \Phi\left(0, \frac{p(1-p)}{n}\right).$$

from I,

$$P\left(|\bar{R}_n - p| > \frac{x \sqrt{p(1-p)}}{\sqrt{n}}\right).$$



$$\therefore P\left(|\bar{R}_n - p| > \frac{x \sqrt{p(1-p)}}{\sqrt{n}}\right) = 2[1 - \phi(x)] \\ \approx 2[1 - \phi(x)]$$

$$\therefore P(|\bar{R}_n - p| > x) \approx 2\left[1 - \phi\left(\frac{\sqrt{n} x}{\sqrt{p(1-p)}}\right)\right]$$

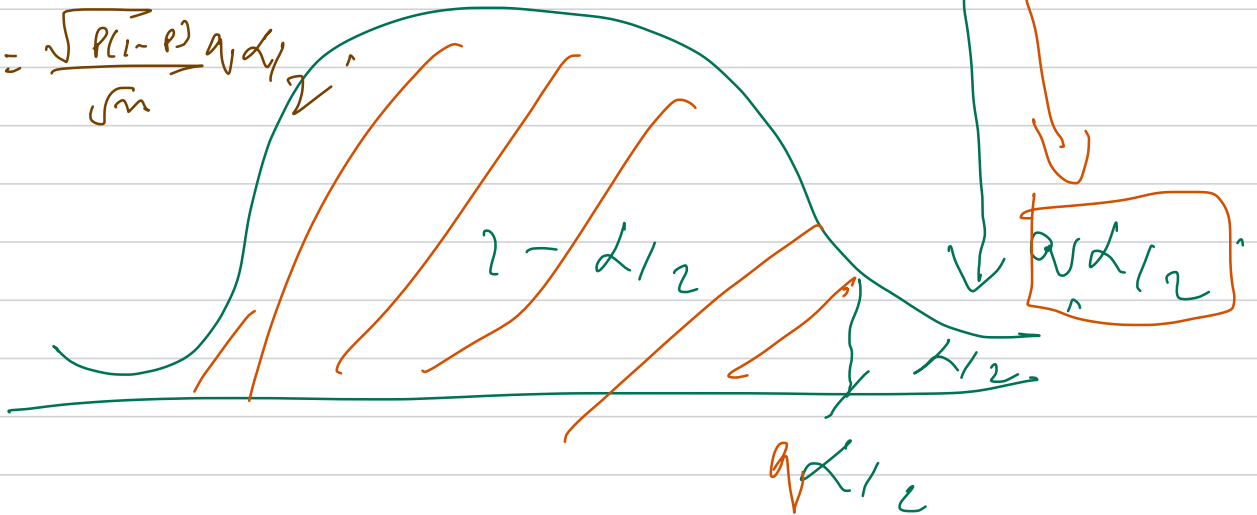
I am looking for z s.t.

$$2 \left(1 - \Phi \left(\frac{z\sqrt{n}}{\sqrt{p(1-p)}} \right) \right) = \alpha.$$

$$\Phi \left(\frac{z\sqrt{n}}{\sqrt{p(1-p)}} \right) = 1 - \alpha/2.$$

$$\frac{z\sqrt{n}}{\sqrt{p(1-p)}} = q_{\alpha/2} \quad \frac{z\sqrt{n}}{\sqrt{p(1-p)}} = \Phi^{-1}(1 - \alpha/2).$$

$$\Rightarrow z = \frac{\sqrt{p(1-p)}}{\sqrt{n}} q_{\alpha/2}.$$



By definition $\boxed{\Phi^{-1}(1 - \alpha/2) = q_{\alpha/2}}.$

$$\Rightarrow \boxed{x = z_{\alpha/2} \frac{\sqrt{p(1-p)}}{\sqrt{n}}}, \quad (\bar{X}_n - p) \leq \frac{z_{\alpha/2} \sqrt{p(1-p)}}{\sqrt{n}}$$

Now, $P[\bar{X}_n - p \leq x] \approx 1 - \alpha$ (1 - α)

$$\Rightarrow P(-x \leq \bar{X}_n - p \leq x) \approx 1 - \alpha.$$

$$\Rightarrow P(\bar{X}_n - x \leq p \leq \bar{X}_n + x) = 1 - \alpha.$$

$$\Rightarrow P\left(\bar{X}_n - \frac{z_{\alpha/2} \sqrt{p(1-p)}}{\sqrt{n}} \leq p \leq \bar{X}_n + \frac{z_{\alpha/2} \sqrt{p(1-p)}}{\sqrt{n}}\right)$$

For plugin replace p by $\hat{p} = \bar{X}_n$.

$$\bar{X}_n - \frac{z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} \leq \bar{X}_n + \frac{z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} = 1 - \alpha.$$

$$\therefore \lim_{n \rightarrow \infty} P\left[\bar{X}_n - \frac{z_{\alpha/2} \sqrt{p(1-p)}}{\sqrt{n}}, \bar{X}_n + \frac{z_{\alpha/2} \sqrt{p(1-p)}}{\sqrt{n}}\right]$$

$$\therefore \text{I converge } \lim_{n \rightarrow \infty} \left(\bar{X}_n - \frac{z_{\alpha/2} \sqrt{p(1-p)}}{\sqrt{n}}, \bar{X}_n + \frac{z_{\alpha/2} \sqrt{p(1-p)}}{\sqrt{n}} \right) = 1 - \alpha.$$

CONSERVATIVE BOUND:

$\Rightarrow$ If I generalise my interval, i.e., make it wider, it will still be at least

$1-\alpha$.

$\uparrow$ It is max
for $n=42$.

Now, $P(1-p) \leq 1/4$ No, matter what.

I am going to make it wider but no more than what's needed.

$$P(|\bar{R}_n - p| \leq \infty) = 1$$

Now, For I_{solve} , $\bar{R}_n$ is our unknown parameter

$$P(|\bar{R}_n - p| \leq \kappa) = 1 - \alpha.$$

Again Re-arranging and we get I_{solve} .

$$\sqrt{n} \frac{\bar{X}_n - p}{\sqrt{p(1-p)}} \xrightarrow[n \rightarrow \infty]{(d)} N(0,1).$$

By Slutsky, CLT, CLT.

Regarding the denominator,

$$\sigma_p = \sqrt{p(1-p)}$$

Now, I know asymptotic variance is a consistent estimator of standard deviation.

$$\text{i.e. } E[S^2] = \sigma_p^2.$$

$$\Rightarrow E[S^2] - E[S]^2 = \sigma_p^2 - \sigma_p.$$

$$\therefore \sigma_p^2 = E[S^2] - E[S]^2$$

$$S = \frac{1}{n-1} \sum (x_i - \bar{x}_n)^2.$$

$$\therefore \hat{p}^2 - \hat{p} = \bar{x}_n^2 - \bar{x}_n$$

Gaussian:

Transformation and Symmetry:

$$X \sim N(2, 2).$$

$$P(X \geq -x) \quad x > 0.$$

Imp:

$$= P\left(\frac{X-2}{\sqrt{2}} \geq \frac{-x-2}{\sqrt{2}}\right)$$

$$= P\left(\frac{X-2}{\sqrt{2}} \leq \frac{x+2}{\sqrt{2}}\right) \quad (\text{by symmetry})$$

$$= \Phi\left(\frac{x+2}{\sqrt{2}}\right) = 1 - \Phi\left(\frac{-x-2}{\sqrt{2}}\right).$$

$$\lim \left(1 + \frac{x}{n}\right)^n = e^x.$$

$$n! = \sqrt{2\pi} e^{-n} x n^{n+1/2} = S(n).$$

$$\frac{n!}{S(n)} \rightarrow 1 \text{ as } n \rightarrow \infty.$$



STIRLING'S APPROXIMATION

$x_1, \dots, x_n$ iid μ, σ^2

$$\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i.$$

$$P[|\bar{x}_n - \mu| \geq \epsilon] \approx 1 - \frac{1}{n}.$$

$$\text{Var}(\bar{x}_n) = \sigma^2/n \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Define $Z_n = \sqrt{n}(\bar{x}_n - \mu).$

$$E[Z_n] = 0 \quad \text{Var}(Z_n) = \sigma^2.$$

$$P(Z_n \leq x) \approx F_0(x).$$

$$\begin{aligned} P(X_n \leq y) &= P(\sqrt{n}(\bar{X}_n - \mu) \leq \sqrt{n}(y - \mu)) \\ &= F_0(\sqrt{n}(y - \mu)). \end{aligned}$$

Convergence in Probability -

$$\{X_n\} \rightarrow X \text{ as } n \rightarrow \infty$$

i.e.

$$X_n \xrightarrow{p} X$$

6.2 Statistics, models and Assumptions;

→ whole Trinity!

① Estimation

② Confidence Intervals

③ Hypothesis Testing.

Model and Assumptions;

Example: Boarding a flight!

HYPOTHESIS TESTING: We will ask a very simple question based on data, is method no.1 better than method number 2?

What is the fastest method to board a plain (R2F) or (Window, Middle Aisle)?

↳ It is basically from an outside to inside method.

↓
In this, board the window passengers first, then middle, then aisle.

Boarding time
in minutes
for 2 airlines
→

	RZF	WILMA
Average (mins)	29.2	15.9
Std. Dev. (mins)	2.1	1.3
Sample Size	72	36

↑
Summary statistics (sufficient) -

Sufficient Statistics: It means they are sufficient
for a particular task.

↑
It is say not enough to determine the 90th
percentile for boarding time. / worst case

MODEL AND ASSUMPTIONS:

▷ $\begin{matrix} X \rightarrow \text{RZF} \\ Y \rightarrow \text{WiZMA} \end{matrix} \rightarrow \text{Two sample Test.}$

▷ Assume $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$.

▷ Sample Size: n m .

$x_1, \dots, x_n$

$y_1, \dots, y_m$.

▷ We further assume the 2 samples are independent
↳ Every single random variable is independent.

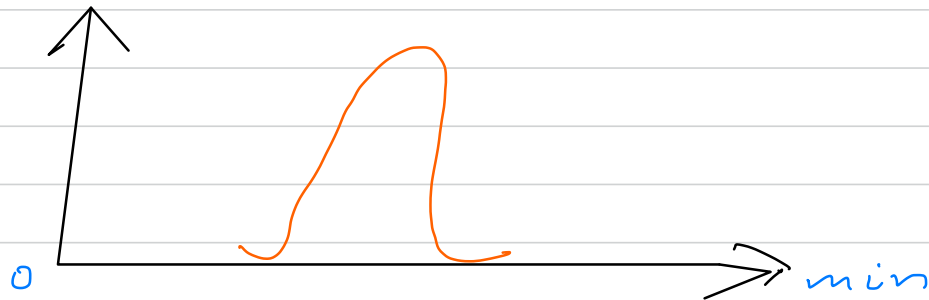
▷ $\rightarrow$ We want to answer this question?

Is $\mu_1 = \mu_2$ or is $\mu_1 > \mu_2$?

▷ By making modelling assumptions, we have reduced the number of ways the hypothesis $\mu_1 = \mu_2$ may be rejected.

We do not allow $\mu_1 < \mu_2$.

→ For this problem, it is a boarding time > 0 .



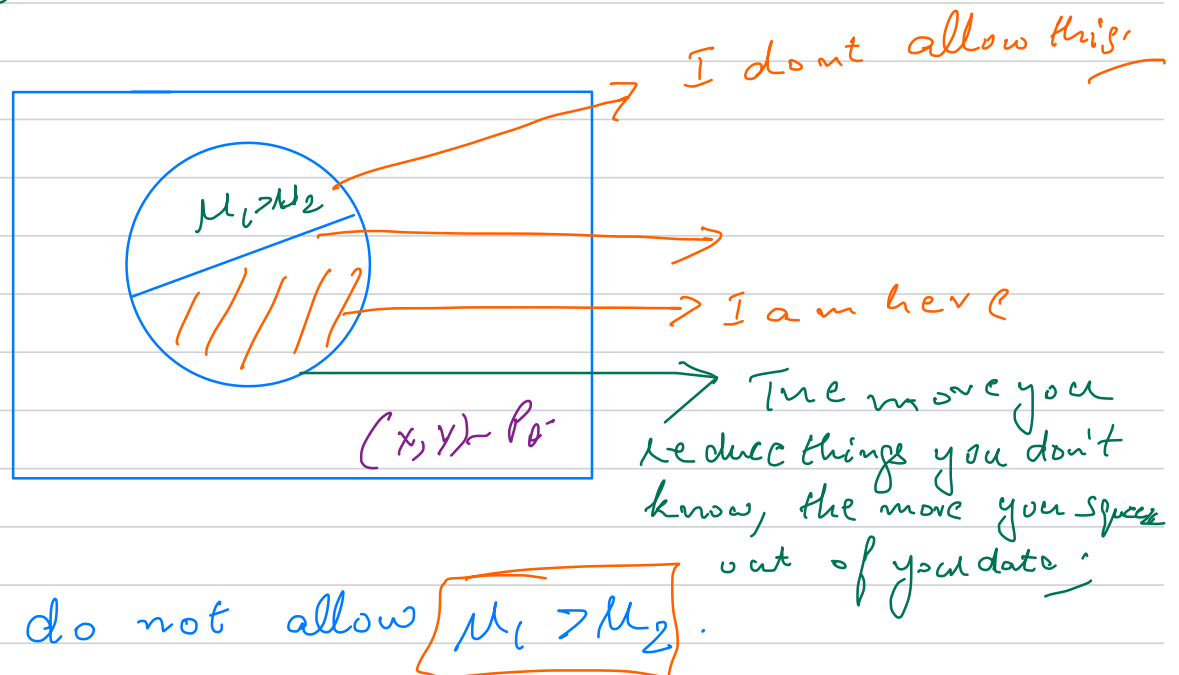
▷ $\mu_1 = \mu_2$ is the a priori / status quo, we assume, before we collect any data.

▷ I am going to start with $\mu_1 = \mu_2$ before I collect any data.

▷ A biological example could be, a bad gene has no particular effect on diabetes.
against
Bad genes cause diabetes.

2.4.8 INTRO TO HYPOTHESIS TESTING:

→ First, I will try to reduce it to a parametric model.



NOTE:

- At this point, I think it means this is my null hypothesis - H_0 .

→ Good scientific inquiry should start from just making this assumption and then collecting data.

If, I would have looked at the summary statistic before, the more likely it would have had been things would have gone your way.

→ But, you should make your assumptions based on your thinking, how you think things actually work!

6.4.4.

→ Two-Sample t-test- one would be used for reference, other thing you are actually trying to test.

→ Examples:

Treatment group v/s control group;
Thing you are actually trying to test, Reference

T-test is just a kind of test.

↑
one type of test.

→

A BETTER HEURISTIC:

→ If this were the case, one was 15.2 and one was 15.1, would that have been enough for me to conclude there is significant difference b/w the two?

→ What if we had a much smaller sample size, would we conclude this from the same margin?

THINGS THAT COME INTO PLAY:

- ① VARIANCE
- ② SAMPLE SIZE
- ③ CONFIDENCE LEVEL
- ④ Fluctuations b/w $\bar{X}_n$ and $\bar{Y}_n$

→ For this example, what we would do is:

→ We are going to remove a little buffer from $\bar{X}_n$ and add a little $\bar{Y}_n$.

The Buffer would depend on the sample size, var etc.

$$\overline{X}_n - \text{Buffer} > \overline{Y}_n + \text{Buffer}_m.$$

And as my sample size goes to ∞ ,
I want no buffer,
They are equal to μ_1 and μ_2 .

→ Using CLT, we are going to talk about
asymptotic distributions, BUT TALKING
ABOUT DISTRIBUTIONS OF $\overline{X}_n$ and $\overline{Y}_n$
IS → ?

ONE SAMPLE TEST:

→ For a one sample test, they give you a benchmark.

Eg: Avg waiting time in ER for a hospital in US is 30 minutes.

→ As a statistician, you collect n -samples at truly random times. Okay, and here I am not making any modelling assumptions I should be making.

Let's just keep it vague for now.

Normal Distribution type.

→ what we want to know,

$$\mu > \textcircled{30} \text{ or not.}$$

May be this was done on a sample of size million. So maybe 30 was the limit of T_m .